Developer task list (update after the strict-LOCO delivery)

P1 = the paper's claims depend on it, **P2** = reviewers asked for it, **P3** = nice to have. The six-classifier results you sent are correct as a pipeline but run on corrupted labels, so everything in B waits for A1 and A2.

Rules for every run

- Strict cross-family LOCO, all 13 folds, seeds 42, 123, 2024, 7, 999, same hyperparameters. Change one thing per run; never tune on the held-out family.
- Same CSV format as `strict_loco_all_6classifiers_raw.csv` (one row per circuit, model, seed), plus a `Config` column for ablations, and the per-gate score files (`Node, y_true, score, pred`).

A. Fix first

☐	**P1**	**A1 Labels.** In the gate-score files, every s35932 circuit has 303–306 positives of which only about 12 have Trojan-like names (`troj*`, `Trigger*`, `tempn*`); 288 are `CRC_OUT_*` outputs and 3 are `test_*` pins. s13207, s1423 and s15850 each have 3 false `test_se/si/so` positives per circuit. About 2,100 of 3,639 Trojan labels are false. Replace the substring filter by an explicit list of Trojan instance names.	Corrected graphs; true Trojan counts per circuit (is s35932 12, 45, or something else?).
☐	**P1**	**A1b Other families.** AES, B19, EthernetMAC10GE and wb-conmax have integer node ids, so names cannot reveal label errors. Check their positives against the Trust-Hub Trojan lists.	One line per family: checked, count.
☐	**P1**	**A2 Features vs. paper.** $f_2 = \lvert TP - TP_{\text{clean}} \rvert$ needs the clean twin of each circuit, so the method is not golden-reference-free. $f_3 = 1 - TP$. Decide: drop f_2 and rerun, or keep it and let the authors change the claims. Give the exact formula of all 8 features.	Decision plus formulas.
☐	**P2**	**A3 Correlation and PCA.** Recompute on the final dataset and state the gate count (the 359,746 in the draft matches nothing: 62 ISCAS+AES circuits have 305,524 nodes, all 86 have 1,434,807).	New S3 and S4 CSVs.
☐	**P2**	**A4 Determinism.** Two identical runs must give identical results (earlier, the `raw_8` baseline differed between runs).	Yes/no, and the cause if no.

B. Reruns after A1 and A2

☐	**P1**	**B1 six classifiers**	GNN head, SVM, XGBoost, CatBoost, RF, ET on the corrected data.	The paper's Tables 2, 3, 5, 6, 7 and Fig. 3 regenerate from it.
☐	**P1**	**B2 ablations**	`raw_8`, `emb_15`, hybrid; no jumping knowledge; no residual; plain cross-entropy; no tree weights; depth 1, 2, 4, 5; drop each feature and the correlated groups; Extra Trees feature importances as a CSV.	Reviewers 1 and 2. The earlier ablation is withdrawn.
☐	**P2**	**B3 cross-fitting**	Trees on out-of-fold embeddings.	Reviewer 1.
☐	**P2**	**B4 stimulus sensitivity**	Extend 1k/10k to the ISCAS, B19, wb-conmax and EthernetMAC10GE circuits and add 100,000 vectors; save the rare-node sets and report their overlap.	Reviewer 1.
☐	**P3**	**B5 prior detector**	One published detector under the same LOCO protocol.	Reviewer 1.
☐	**P3**	**B6 c6288 TP**	Benign vs. Trojan TP histograms per family and the share of benign gates with $TP < 0.1$.	Reviewer 1.
☐	**P3**	**B7 ranking, thresholds**	Gates inspected to find all Trojan gates, from the saved scores; optionally a threshold chosen on training folds only.	Reviewers 1 and 2.

C. Figures, references, release

☐	**P2**	**C1 Fig. 2 (`htpipe.pdf`).** Show the 8 features entering GraphSAGE; remove result numbers.	New `htpipe.pdf`.
☐	**P3**	**C2 References.** JEIT issue number and end page (Crossref, DOI 10.11999/JEIT221201); optionally ISCAS-85/89 original papers and a Trust-Hub accessed date. Do *not* use the BibTeX block in your document: it contradicts the IEEE Xplore and ScienceDirect records.	Details.
☐	**P3**	**C3 Release.** Public repository with code, graphs, corrected labels, per-circuit results, gate scores.	URL.

Send back

New strict-LOCO CSVs and gate scores; written answers to A1–A4; the ablation CSVs; the figure and BibTeX details.